



TITLE: CA1 ADVANCED PROGRAMMING TECHNIQUES

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MODULE NAME: ADVANCED PROGRAMMING TECHNIQUES

MODULE CODE: B9CY108

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1.PART 1 – C# PROGRAMMING:

1.1 QUESTION 1: CONTACT BOOK APPLICATION

FILE STRUCTURE AND CODE DESCRIPTION:

I organized my Visual Studio .NET Console Application project named Ques1 by keeping Contact.cs for data structure and procedures and Program.cs for core logic and menu-driven interface in a single folder.

1.Contact.cs

The ContactBook class keeps all the details for each contact organized. It checks that mobile numbers are valid, can show full contact information, and easily converts data for saving or loading. It also gives a simple summary, so contacts are easy to browse.

2.Program.cs

This Program.cs file is the main engine of the contact book. It loads saved contacts, lets you add, view, edit, or delete anyone, checks mobile numbers, and keeps everything running smoothly until you choose to exit.

Result:

```
PS C:\Users\Shreya\CA1 PROGRAMMING> & 'c:\Users\Shreya\.vscode\extension
.11-win32-x64\.debugger\x86_64\vsdbg.exe' '--interpreter=vscode' '--conn
1776393c3'
Main Menu
1: Add Contact
2: Show All Contacts
3: Show Contact Details
4: Update Contact
5: Delete Contact
0: Exit
Enter choice: 1

Enter details for Contact #1:
First Name: Shreya
Last Name: Shinde
Company: Dbs
Email: shreya@mail.com
Birth Date (dd/MM/yyyy): 090204
Mobile Number (9 digits, cannot start with 0): 123456789
Contact added.

Main Menu
1: Add Contact
2: Show All Contacts
3: Show Contact Details
4: Update Contact
5: Delete Contact
0: Exit
Enter choice: 1

Enter details for Contact #2:
First Name: Riya
Last Name: Shinde
Company: SAGHS
Email: riya@mail.com
Birth Date (dd/MM/yyyy): 031108
Mobile Number (9 digits, cannot start with 0): 234567891
Contact added.
```

```
PS C:\Users\Shreya\CA1 PROGRAMMING> & 'c:\Users\Shreya\.vscode\extensions\ms-dotn  
.11-win32-x64\.debugger\x86_64\vsdbg.exe' '--interpreter=vscode' '--connection=3f7  
1776393c3'
```

Enter details for Contact #20:

First Name: Rugved

Last Name: Latke

Company: pan

Email: rugved@mail.com

Birth Date (dd/MM/yyyy): 150403

Mobile Number (9 digits, cannot start with 0): 673254877

Contact added.

Main Menu

1: Add Contact

2: Show All Contacts

3: Show Contact Details

4: Update Contact

5: Delete Contact

0: Exit

Enter choice: 1

Contact book full (maximum 20 contacts).

Main Menu

1: Add Contact

2: Show All Contacts

3: Show Contact Details

4: Update Contact

5: Delete Contact

0: Exit

Enter choice: 2

--- All Contacts ---

1. Shreya Shinde - 123456789

2. Riya Shinde - 234567891

3. Tanvi Gaikwad - 345672886

4. Parimita Gaikwad - 456789123

5. Hemant Gaikwad - 567891234

6. Anand Gaikwad - 678912345

7. Shweta Gaikwad - 789123456

8. Shakuntala Gaikwad - 891234567

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Shreya\CA1 PROGRAMMING> & 'c:\Users\Shreya\.vscode\extensions\ms-dotnettools.csharp-1.11-win32-x64\.debugger\x86_64\vsdbg.exe' '--interpreter=vscode' '--connection=3f7dff73e68647a1776393c3'

--- All Contacts ---
1. Shreya Shinde - 123456789
2. Riya Shinde - 234567891
3. Tanvi Gaikwad - 345672886
4. Parimita Gaikwad - 456789123
5. Hemant Gaikwad - 567891234
6. Anand Gaikwad - 678912345
7. Shweta Gaikwad - 789123456
8. Shakuntala Gaikwad - 891234567
9. Ramchandra Gaikwad - 901234567
10. Bhimrao Shinde - 907856342
11. Jyashree Shinde - 346712869
12. Pravin Shinde - 784523896
13. Pranali Shinde - 563725892
14. Shalini Gaikwad - 563413289
15. Nlaini Pokhare - 126783594
16. Rani Pokhare - 762534891
17. Sunny Pokhare - 333467832
18. Rajshree Giakwad - 457583921
19. Genbhau Shinde - 652387468
20. Rugved Latke - 673254877

Main Menu
1: Add Contact
2: Show All Contacts
3: Show Contact Details
4: Update Contact
5: Delete Contact
0: Exit
Enter choice: 3
Find by (1) index or (2) mobile? Enter 1 or 2: 1
Enter contact index (1-based): 18

--- Contact Details ---
First Name : Rajshree
Last Name  : Giakwad
Company    : MUM
```

```
1776393c3'
20. Rugved Latke - 673254877

Main Menu
1: Add Contact
2: Show All Contacts
3: Show Contact Details
4: Update Contact
5: Delete Contact
0: Exit
Enter choice: 3
Find by (1) index or (2) mobile? Enter 1 or 2: 1
Enter contact index (1-based): 18

--- Contact Details ---
First Name : Rajshree
Last Name  : Giakwad
Company    : MUM
Mobile No. : 457583921
Email      : rajshree@mail.com
Birth Date : 010173

Main Menu
1: Add Contact
2: Show All Contacts
3: Show Contact Details
4: Update Contact
5: Delete Contact
0: Exit
Enter choice: █
```

Added, updated, displayed, validated, and listed 20 contacts.

2.PART 1 – C# PROGRAMMING:

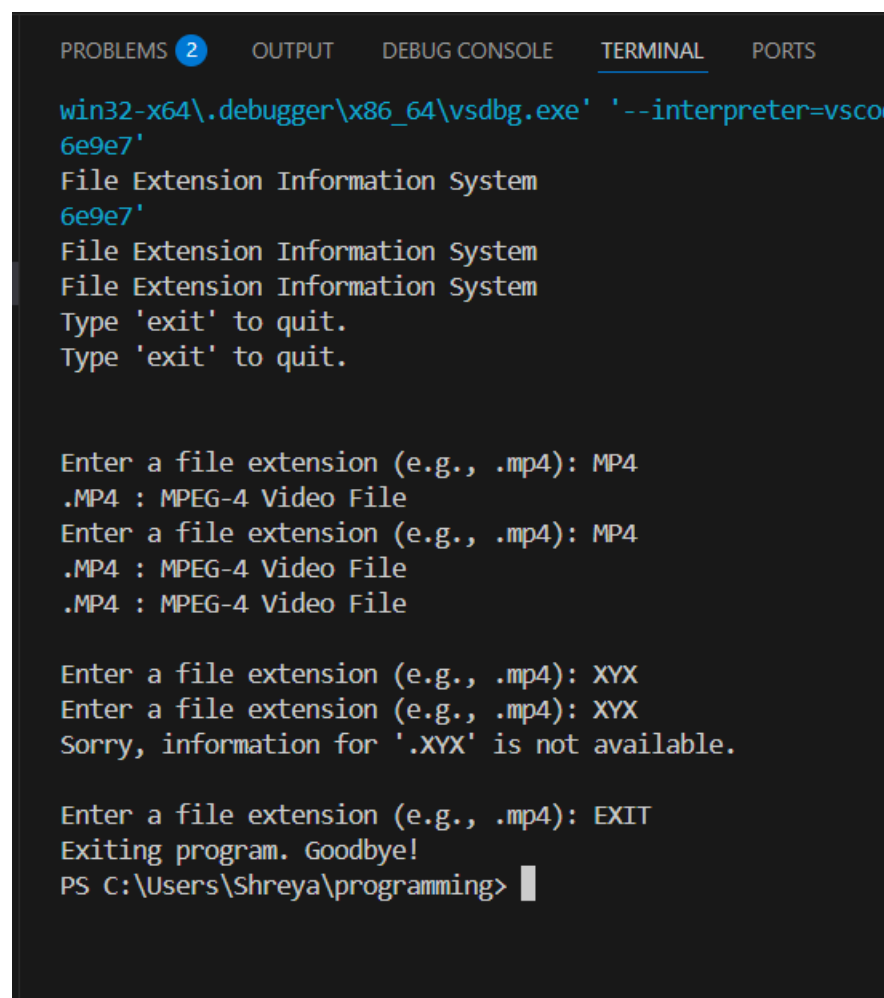
2.1 QUESTION 2: CONTACT BOOK APPLICATION

FILE STRUCTURE AND CODE DESCRIPTION:

C# program provides information for 20+ file extensions, uses appropriate data structures, and handles unexpected inquiries gracefully

The FileExtensionInfo C# program defines a class containing the Main method, using a case-insensitive dictionary to store 20+ file extensions with their descriptions. It continuously prompts the user for input, normalizes entries by adding a dot if missing, looks up descriptions, handles unknown or invalid queries gracefully, and exits when "exit" is typed, demonstrating OOP, loops, conditionals, and efficient key-value storage.

Result:



```
win32-x64\.debugger\x86_64\vsdbg.exe' '--interpreter=vscode
6e9e7'
File Extension Information System
6e9e7'
File Extension Information System
File Extension Information System
Type 'exit' to quit.
Type 'exit' to quit.

Enter a file extension (e.g., .mp4): MP4
.MP4 : MPEG-4 Video File
Enter a file extension (e.g., .mp4): MP4
.MP4 : MPEG-4 Video File
.MP4 : MPEG-4 Video File

Enter a file extension (e.g., .mp4): XYX
Enter a file extension (e.g., .mp4): XYX
Sorry, information for '.XYX' is not available.

Enter a file extension (e.g., .mp4): EXIT
Exiting program. Goodbye!
PS C:\Users\Shreya\programming>
```


The program lets you check file extensions, recognizes popular ones like MP4, warns you of unknown ones like XYZ, and ends neatly when you hit "exit."

3.PART 2 – PYTHON PROGRAMMING:

3.1 QUESTION 3: TCP-BASED STUDENT APPLICATION SYSTEM

FILE STRUCTURE AND CODE DESCRIPTION:

In Visual Studio, create a new Python project, add a Python file, give it a name, and start coding.

Ques3_client.py:

As a helpful assistant, this program assists students in completing an admission form. It gracefully compresses and sends your answers to the server after requesting your name, residence, education, course, and start date. After being sent, it waits for a response to let you know if your application was accepted or if there was a problem. The main () function acts as the conductor, coordinating the queries, sending your data, and presenting the server's response in an understandable way.

Ques3_server.py:

It safely stores the application data in a database after waiting for students to submit it. Each application is examined to make sure the dates are correct; the course is authentic, and all fields are filled out. If everything goes as planned, the student is given a unique application number. The server can communicate with multiple students at once, guarantees a seamless shutdown, and gently resolves issues.

Result:

```
PS C:\Users\Shreya\20074879\Ques 3> python Ques3_server.py
Server listening on 127.0.0.1:5000 ... (CTRL+C to stop)
Accepted connection from ('127.0.0.1', 60572)
[('127.0.0.1', 60572)] Stored application B728188B for Shreya Shinde (MSc Cyber Security)
```

```
PS C:\Users\Shreya\20074879\Ques 3> python Ques3_client.py
=== DBS Admission Application ===
Enter your full name: Shreya Shinde
Enter your address: Mumbai
Enter your educational qualifications: Btech
Available courses: MSc Cyber Security, MSc Information Systems & Computing, MSc Data Analytics
Enter the course you wish to enroll in: MSc Cyber Security
Enter intended start year (YYYY): 2025
Enter intended start month (MM or M): 09

Application successful.
Your application number: B728188B
PS C:\Users\Shreya\20074879\Ques 3>
```

The MSc Cyber Security program, which will start in September 2025, accepted Shreya Shinde's application. The server saved her data and gave her application number B728188B.

4.PART 2 – PYTHON PROGRAMMING:

4.1 QUESTION 4: WEB SCRAPING PROJECT

FILE STRUCTURE AND CODE DESCRIPTION:

Ques4_program.py:

As a useful tool, this software collects hotel room rates from two websites for a chosen period. It visits the sites, gathers hotel names, room types, and rates, and kindly confirms if scraping is allowed. It makes sure that every site has ten or more entries, stores everything in a CSV file, and adds extra details like the URL and timestamp. Finally, the information is shown on the screen, so you can quickly compare prices and decide which option is best.

Result:

```
PS C:\Users\Shreya\20074879> & C:/Users/Shreya/AppData/Local/Microsoft/WindowsApps/python
3.11.exe "c:/Users/Shreya/20074879/Ques 4/Ques_program.py"

--- Scraping https://hotel1.tiiny.site/ ---
Extracted 35 rows (after ensuring minimum).

--- Scraping https://booking-hotels2.tiiny.site/ ---
Extracted 35 rows (after ensuring minimum).

Saved 70 rows to hotels_combined.csv

site,hotel,room name,price_raw,checkin,checkout,scraped_at
https://hotel1.tiiny.site/,Clayton Hotel Burlington Road,Standard King Room • 1 king
€ 143,2025-12-20,2025-12-30,2025-12-08T08:11:33.134669
https://hotel1.tiiny.site/,The Chancery Hotel,Standard King Room • 1 queen bed,€ 234,
-12-20,2025-12-30,2025-12-08T08:11:33.134669
https://hotel1.tiiny.site/,Beckett Locke,One-Bedroom Apartment • 1 bedroom • 1 living
m • 1 queen bed,€ 190,2025-12-20,2025-12-30,2025-12-08T08:11:33.134669
```

Hotel room rates were collected by the scraper from two websites. Each site had 35 rooms, for a total of 70 entries. The CSV file hotels_combined.csv, which includes all the information, including hotel names, room types, and prices, makes it simple for travelers to compare options and choose the perfect stay.

Git Repository Link: https://github.com/shinde-bot/ca1_programming.git